Module 1 - Cloud Foundations, Storage, Serverless & CI/CD

- **AWS Services Covered** - AWS IAM, AWS S3, AWS S3 Tables, AWS Lambda, AWS CodeBuild, GitHub, GitHub Actions, AWS SNS

✅ Class - 1

- Importance of Cloud
- Cloud vs On-Premises Architectures
- AWS vs Azure vs GCP (Data Engineering perspective)
- AWS Free Tier Account Setup
- IAM Users, Roles & Policies
- AWS CLI Setup and Authentication
- AWS S3 Fundamentals
- Create S3 Buckets, Folders & Objects
- Access S3 using AWS CLI
- AWS S3 Tables – managed tabular data on S3 (modern lakehouse concept)

✅ Class - 2

- AWS Lambda Fundamentals
- AWS Lambda Architecture & Execution Model
- Writing and Deploying Lambda Functions
- S3 Event Notification Triggers
- Lambda Testing using Test Events
- Lambda Packaging without External Dependencies
- Lambda Layers – concept & usage
- Packaging and Deployment using Layers

✅ Class - 3

- Lambda + CI/CD + Messaging
- Lambda with External Python Dependencies
- Lambda Triggers Overview
- S3 Create/Update/Delete Object Triggers

- Reading CSV Data from S3 in Lambda
- CI/CD for Serverless Applications
- AWS CodeBuild Setup & GitHub Integration
- Lambda CI/CD Pipeline using CodeBuild
- AWS SNS Fundamentals
- SNS Topics & Subscriptions
- Email Subscriptions
- Publishing S3 Events to SNS
- Sending SNS Messages from Lambda using boto3

Module 2 - Compute, Messaging, Orchestration & Data Ingestion

- **AWS Services Covered** - AWS EC2, AWS SQS, AWS EventBridge, EventBridge Pipes, EventBridge Scheduler, AWS RDS, AWS Secrets Manager, AWS Glue, AWS AppFlow

✅ Class - 1

- Compute, Queues & Eventing
- AWS EC2 Fundamentals
- EC2 Instance Types & Pricing
- EC2 Setup & SSH Access
- AWS SQS Fundamentals
- SQS vs Kafka (use-case driven comparison)
- Standard vs FIFO Queues
- Sending & Receiving Messages
- Publishing Bulk Messages to SQS using Lambda
- SQS Triggered Lambda Processing
- AWS EventBridge Fundamentals
- EventBridge Pipes (SQS → Lambda)
- EventBridge Scheduler for Time-based Pipelines

✅ Class - 2

- Databases, Metadata & ETL
- AWS RDS Fundamentals
- MySQL Setup on AWS RDS

- Secrets Management using AWS Secrets Manager
- MySQL Access via Terminal & Python
- AWS Glue Overview & Architecture
- Glue Data Catalog & Metadata Management
- Glue Crawlers & Connections
- Crawling CSV & Partitioned Data in S3
- Crawling RDS Tables
- Incremental Loads using Glue Job Bookmarks
- S3 → MySQL ETL using Glue Visual ETL
- AWS AppFlow – ingesting SaaS data into S3 / RDS

Module 3 - Analytics, Streaming & Big Data Processing

- **AWS Services Covered** - AWS Athena, AWS Redshift, Amazon MWAA (Managed Airflow), AWS EMR, AWS Kinesis, Kinesis Streams, DynamoDB, DynamoDB Streams, Kinesis Firehose

✅ Class - 1

- Query Engines & Data Warehousing
- AWS Athena Architecture
- Athena vs Spark – decision criteria
- Querying S3 Data using Glue Catalog
- External Tables in Athena
- Athena Queries from Lambda
- JDBC & Federated Queries using Lambda
- AWS Redshift Architecture
- Redshift Cluster Setup
- Redshift Spectrum
- COPY Command for Data Ingestion
- UNLOAD Data to S3 (single & multiple files)
- Load/Unload using Manifest Files

✅ Class - 2

- Orchestration & Big Data Processing
- Apache Airflow Overview & Architecture

- Complete Airflow Setup on AWS (MWAA)
- Airflow UI Walkthrough
- DAG Creation (Sequential & Parallel Tasks)
- AWS EMR Architecture
- EMR Cluster Setup
- EMR Proxy & Tunnel Management
- Hadoop, Hive & Spark Access
- Airflow EMRStepOperator
- Spark Pipeline on Food Inspection Dataset
- Hive Querying via HUE UI

✅ Class - 3

- Streaming & NoSQL
- AWS Kinesis Fundamentals
- Kinesis vs Kafka
- DynamoDB Architecture
- DynamoDB & DynamoDB Streams Setup
- CDC Capture via DynamoDB Streams
- EventBridge Pipes for CDC Events
- Kinesis Firehose Setup
- Firehose → S3 Data Delivery
- Industry Project - 1: Gadget Sales Forecasting & Analytics
- Tech Stack: DynamoDB Streams, EventBridge Pipes, Kinesis, Firehose, S3, Athena

Module 4 - Workflow Orchestration & Event-Driven Systems

- **AWS Services Covered** - AWS Step Functions

✅ Class - 1

- Step Functions & Event-Driven Architecture
- AWS Step Functions Overview
- State Machines & Key Components
- Creating Step Functions

- Lambda-to-Lambda Orchestration
- Error Handling & Retries
- Industry Project 2 : Event-Driven Sales Data Analysis
- Tech Stack: S3, EventBridge Rules, Step Functions, Lambda, SQS, DynamoDB

Module 5 - Advanced Data Engineering & AI Exposure

- **AWS Services Covered** - AWS DMS, Amazon Bedrock, Amazon Q

✅ Class - 1

- AWS DMS for CDC Pipelines
- Full Load + CDC from RDS to S3 / Redshift
- Designing Low-Latency CDC Architectures
- Amazon Bedrock – AI models for data workflows
- AI-driven enrichment & summarization use cases
- Amazon Q – AI-assisted data & cloud operations
- How AI fits into modern Data Engineering pipelines

Module 6 - Industrial Projects

- AWS Services Covered
- S3, Lambda, IAM, CLOUDWATCH, EC2, SNS, SQS
- Event Bridge Scheduler, Event Bridge Pipe, Kinesis, Kinesis Firehose, DynamoDB, SNS, SQS
- Step Function, EMR, GLUE, RDS, ATHENA, REDSHIFT, AWS Code Build For CICD
- AWS S3 Tables, AWS DMS, AWS AppFlow, Amazon Bedrock, Amazon Q
- Industry Project - 3 : Incremental ingestion & analysis of Airlines data
- Industry Project - 4 : Weather data analysis using weather api
- Industry Project - 5 : Quality Movie Data Analysis
- Industry Project - 6 : Real Time Food Delivery Analysis (CICD Pipeline)
- Exclusive Projects Of Data Engineering with AI, LLMS & AI Agents

Your syllabus looks **very strong and industry-oriented** for **AWS Data Engineering**. Since you already have **4 years of AI/ML/LLM experience at TCS**, this stack can **help you shift toward Data Engineering + AI pipelines**, which is a **high-paying domain in India (20–40 LPA range)**.

Let me break it down **practically**.

1️⃣ Overall Evaluation of This Curriculum

This course covers **almost the full AWS Data Engineering ecosystem**.

✓ Core Data Engineering

- **Storage & Lakehouse** → S3, S3 Tables
- **Compute** → EC2, Lambda
- **Data Processing** → Glue, EMR, Spark
- **Data Warehouse** → Redshift
- **Query Engine** → Athena

✓ Event Driven Systems

- EventBridge
- SQS
- SNS
- Step Functions

✓ Streaming / Real-time

- Kinesis
- DynamoDB Streams
- Firehose

✓ Data Ingestion

- AppFlow
- DMS (CDC pipelines)

✓ Orchestration

- Airflow (MWAA)

✓ DevOps

- GitHub Actions
- CodeBuild
- CI/CD pipelines

✓ **AI Integration**

- Amazon Bedrock
- Amazon Q

👉 This is **very close to a real production architecture**.

2️⃣ Industry Architecture You Will Learn

This syllabus basically teaches a **Modern Data Platform**.

Example pipeline:

```
API / SaaS / DB
   |
   ▼
AppFlow / DMS / Kinesis
   |
   ▼
S3 Data Lake
   |
   ▼
Glue ETL / EMR Spark
   |
   ▼
Redshift / Athena
   |
   ▼
Dashboards / ML Models
```

And **event-driven pipeline**:

```
S3 Upload
  |
  ▼
EventBridge
  |
  ▼
```

Lambda

|

▼

SQS / DynamoDB

|

▼

Athena / Redshift

3️⃣ Very Strong Topics in This Course

These are **high demand in the market**.

⭐ **Serverless Data Engineering**

- Lambda
- S3
- Athena
- EventBridge
- Step Functions

⭐ **Streaming Systems**

- Kinesis
- DynamoDB Streams
- Firehose

⭐ **Big Data Processing**

- EMR
- Spark
- Hive

⭐ **Orchestration**

- Airflow (MWAA)

⭐ **Modern Lakehouse**

- S3 Tables
- Athena
- Glue Catalog

⭐ **CDC Pipelines**

- AWS DMS
- DynamoDB Streams

⭐ **AI Integration**

- Bedrock
- AI-driven pipelines

This is **very modern architecture (2024-2026 level)**.

4️⃣ Things Missing (Minor)

For **top product companies**, these tools are also useful:

- **Apache Kafka**
- **Databricks**
- **Delta Lake / Iceberg**
- **dbt**
- **Terraform**

But for **AWS Data Engineer roles**, this course is already **80–85% sufficient**.

5️⃣ Salary Potential After This Stack (India)

Approx salary if you master these tools:

Experience	Salary
4 years	18–28 LPA
5 years	25–35 LPA
Senior DE	35–50 LPA

Since you already have **AI + LLM experience**, your profile can become:

AI Data Engineer / ML Data Platform Engineer

These roles pay **even higher**.

6️⃣ Real Companies Using Similar Architecture

Many companies use similar stacks:

- Amazon
- Flipkart

- Uber
- Swiggy
- Zomato
- Netflix

Typical stack there:

S3 + Spark + Airflow + Kafka + Redshift

Your syllabus is **very close to that**.

7️⃣ Biggest Advantage for YOU

Since you already work with:

- AI
- NLP
- LLM
- RAG

You can build **AI Data Pipelines**.

Example:

Raw Data

|

▼

S3

|

▼

Glue ETL

|

▼

Vector DB

|

▼

RAG Pipeline

|

▼

LLM (Bedrock)

This is **next-generation Data Engineering**.

8️⃣ My Honest Opinion

If this course actually teaches **hands-on projects**, then it is:

⭐ **8.5 / 10 course**

Very strong for:

- AWS Data Engineer
- Data Platform Engineer
- AI Data Engineer

✅ My suggestion for you

Focus on these modules most:

1️⃣ **Glue + Athena + S3**
2️⃣ **Kinesis Streaming**
3️⃣ **Airflow orchestration**
4️⃣ **Lambda pipelines**
5️⃣ **Redshift warehouse**